FEDERAL SPECIFICATION

CANNULA, LARYNGEAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers cannula, laryngeal for suction and removal of body secretions and aspiration of bronchial tract.

1.2 Classification.

1.2.1 *Types.*—Cannulae shall be of the following types, as specified (see 6.1):

Type I.—Abraham design (flexible tube).

Type II.—Jackson design (aspirating tube).

Type III.—Yankauer design (suction tube).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG-I-526 — Instruments, Dental and Surgical; General Specifications.

QQ-B-626 — Brass, Leaded and Non-Leaded; Rods, Shapes, Forgings and Flat Products with Finished Edges (Bars, Flat Wire and Strips).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-P-12902—Packing of Surgical Dental and Veterinary Instruments, Diagnostic Equipment, and Related Items.

Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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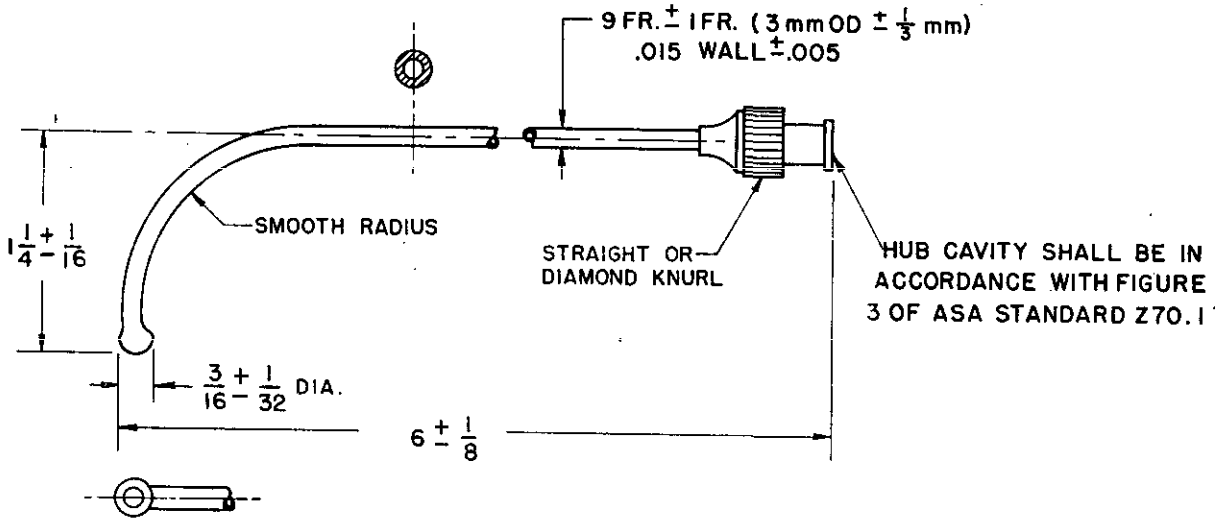
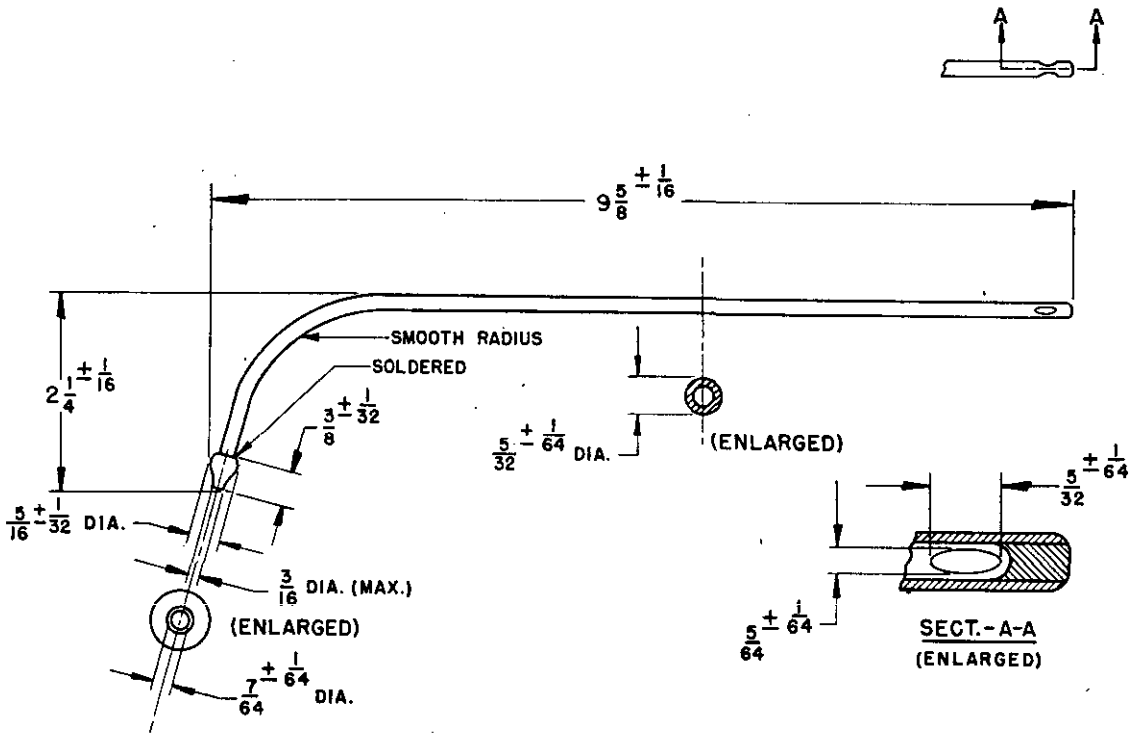


FIGURE 1.—Type I, Abraham design (flexible tube).



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FIGURE 2.—Type II, Jackson design (aspirating tube).

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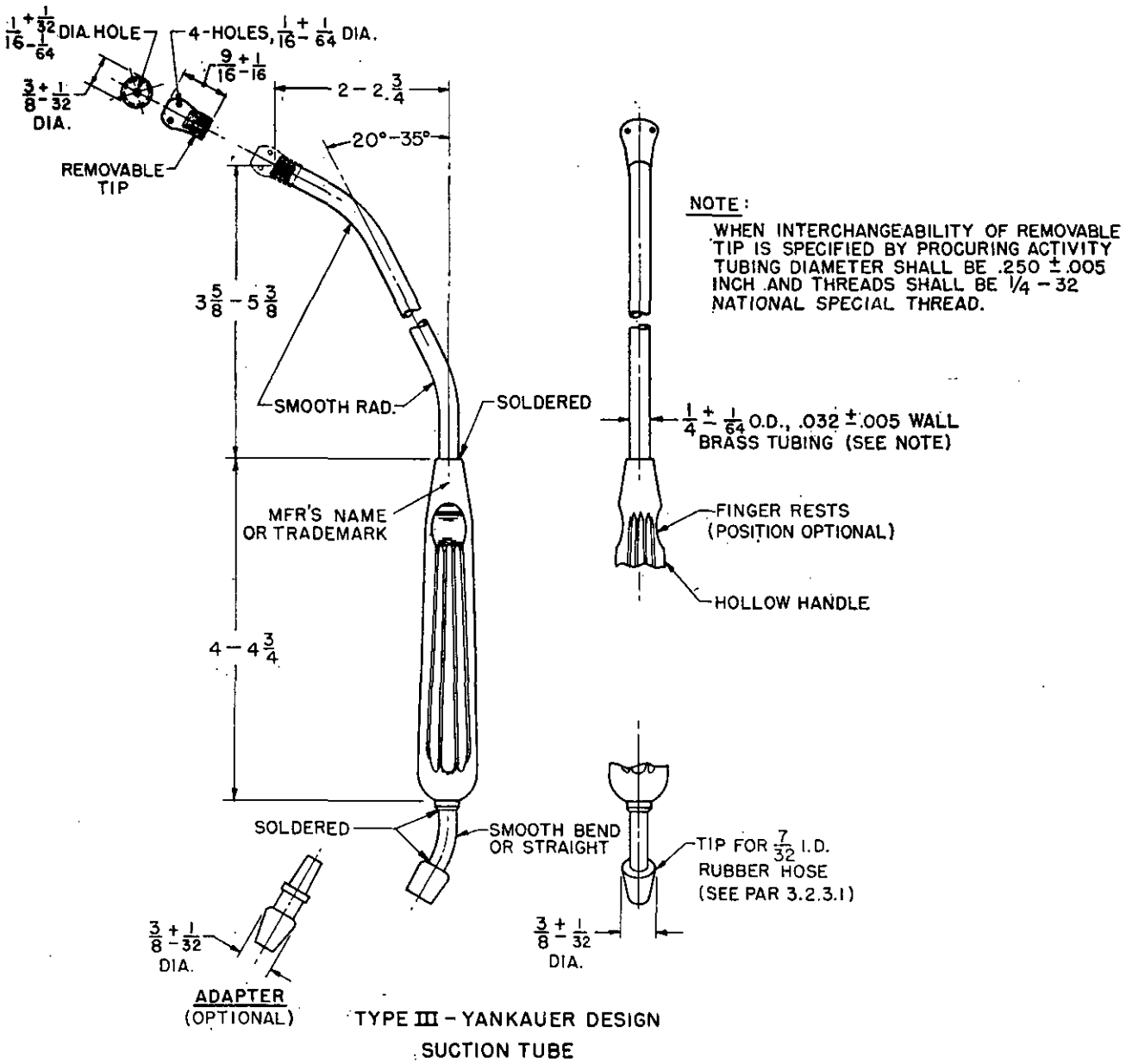


FIGURE 3.—Type III, Yankauer design (suction tube).

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TABLE I.

Type	Component	Material		
		Specification	Class	Composition
I (Abraham)	Cannula	GG–I–526	4	P*
	Hub	GG–I–526	4	N**
	or			
	Hub, alternate material	QQ–B–626		22**
II (Jackson)	Entire instrument	GG–I–526	4	N
III (Yankauer)	Entire instrument	GG–I–526	4	N

* Nickel plated only.

** Nickel or chromium over nickel plated in conformance with plating requirements and corrosion tests requirements for class 4, composition N material of Federal Specification GG–I–526.

American Standards Association Inc. Standard:

Z70.1–1955 (UDC615.17) — American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Applications for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Material.—Cannulae shall be fabricated of classes and compositions, as specified in table I, in accordance with Federal Specification GG–I–526 and QQ–B–626, as applicable:

3.2 Style, design, and dimensions.

3.2.1 *Type I (Abraham).*—Type I shall be a curved flexible laryngeal cannula, Abraham design, with bulbous tip and Luer lock hub. Cannula shall be in accordance with figure 1.

3.2.1.1 *Hub.*—Hub shall be a Luer lock type with hub cavity and flange in accordance with figure 3 of American Standard Association (ASA) Standard Z70.1–1955, UDC 615.47. Hub cavity shall be well plated down to the entrance of the lumen.

3.2.1.1.1 *Diameter and depth of hub cavity.*—When tested as specified in 4.4.1, the hub shall fit the plug gage within the "GO"–"NO GO" limits, and there shall be no movement of the hub on the gage.

3.2.1.1.2 *Freedom from foreign matter.*—When tested as specified in 4.4.2 the hub cavity shall show complete freedom from metal chips, peeled plating, and other foreign matter.

3.2.1.1.3 *Fluid tightness.*—When tested as specified in 4.4.3, the connection of the cannula proper with the hub shall show no leakage, and the connection of the hub to the plug gage shall not show leakage exceeding the formation of a falling drop.

3.2.1.1.4 *Security of assembly.* — When tested as specified in 4.4.4, the cannula and hub shall not loosen or become detached.

3.2.2 *Type II (Jackson).*—Type II shall be an aspirating tube, velvet eye type, Jackson design, in accordance with figure 2. Cannula shall be suitable for aspiration of bronchial tract.

3.2.3 *Type III (Yankauer).*—Type III shall be a suction tube, Yankauer design, with removable tip, suitable for suction and removal of body secretions. Cannula shall be in accordance with figure 3.

3.2.3.1 *Hose connecting tip.*—The proximal end of the suction tube shall be provided with

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a suitable tip, as shown in figure 3, for connecting $\frac{7}{32}$-inch (inside diameter) rubber tubing. As an alternate, this may be accomplished with the use of an adapter constructed as shown in figure 3. The adapter shall make a leakproof (slip) fit with the proximal end of the suction tube, and when tested as specified in 4.4.5, leakage shall not exceed the formation of a falling drop.

3.2.3.2 *Removable tip.*—Removable tip shall be as shown in figure 3. When interchangeable removable tip is specified by procuring activity, outside diameter of brass tube shall be 0.250 ± .005 inch and threads shall be $\frac{1}{4}$–32 National Special Thread (see 6.1).

3.3 Finish.—Finish shall be mirror, or mirror and satin.

3.4 Additional requirements. — Except as specified herein, the cannula shall comply with requirements of Federal Specification GG–I–526.

3.5 Workmanship.—Workmanship shall be first class throughout. Cannulae shall be free from defects which detract from their appearance or impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 *For examination.*—Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105.

4.2.2 *For tests.*—Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105. Requirement applicable for individual unit only.

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL
Cannula	Security of assembly	3.2.1.1.4	4.4.4	1-Cannula	L5	1.5
	Diameter and depth	3.2.1.1.1	4.4.1	1-Cannula	II	1.0
	Freedom from foreign matter	3.2.1.1.2	4.4.2	1-Cannula	II	2.5
	Fluid tightness	3.2.1.1.3	4.4.3	1-Cannula	L5	1.5
	Boil and salt spray tests	GG–I–526	GG–I–526	1-Cannula	L2*	**
Cannula and adapter	Adapter leakage	3.2.3.1	4.4.5	1-each	L5	1.5

*A minimum of two samples shall be tested regardless of lot size.

**Acceptance number shall be zero for all sample sizes.

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4.2.3 *Raw materials.*—Four ounces of each lot of each metal used in the construction of the cannula shall be selected for chemical analysis.

4.2.4 Where possible, the same sample unit shall be used for the determination of two or more tests characteristics.

4.3 **Examination.**—The cannula shall be examined to determine compliance with all requirements contained in this specification.

4.4 **Tests.**—Tests shall be conducted as necessary, to determine compliance with specification requirements.

4.4.1 *Diameter and depth.* — Hub cavity shall be tested on a male luer taper gage in accordance with plug gage of figure 5 of ASA Standard Z70.1–1955 (except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.3). The hub shall be slipped onto this gage with light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figures 5 and 6 of ASA Standard Z70.1–1955. In addition, when the hub is fitted onto the gage as specified above, any movement of the hub on the gage, when the hub is pressed lightly from side to side in all directions, shall be noted.

4.4.2 *Freedom from foreign matter.* — The hub cavity shall be visually examined, under good illumination, for metal chips and other foreign matter. In addition, a cotton swab tightly wrapped on the end of a small diameter wood applicator shall be vigorously turned down into the bottom of the hub cavity. When withdrawn, the cotton swab shall be examined, under good illumination, for metal chips and other foreign matter.

4.4.3 *Fluid tightness test.*—With the cannula in a horizontal position, the hub shall be attached, by means of the plug gage described in 4.4.1, to a water-pressure source having a pressure gage. Water shall be run through the cannula so as to fill the hub cavity and the lumen of the cannula. The lumen shall then be sealed and a hydrostatic pressure of 45 pounds per square inch gage (p.s.i.g.) applied for ten (10) seconds.

4.4.4 *Security of assembly test.*—With the cannula securely clamped, the hub shall be submitted to a force (pull) of 25 pounds.

4.4.5 *Adapter leakage test.*—With the cannula in a horizontal position (and the adapter assembled in position) the cannula shall be connected through the adapter to a water pressure source having a pressure gage. Water shall be run through the entire cannula so as to fill the inside of the cannula. The cannula tip (distal end) shall then be sealed and a hydrostatic pressure of 10 p.s.i.g. applied for ten seconds.

5. PREPARATION FOR DELIVERY

The levels of preservation, packaging, and packing used are defined in Federal Standard No. 102.

5.1 **Packaging and packing.**

5.1.1 *Unit of issue.*—One cannula, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table II shall be packaged in each unit, intermediate and exterior container, as applicable for the required level of protection specified in the procurement document.

TABLE II.—*Packaging quantities*

Type	Unit package	Intermediate package	Exterior container
I	1	12	432
II	1	12	360
III	1	6	288

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.1.3 *Level A.* — Packaging and packing shall conform to the requirements of Military Specification MIL–P–12902, Procedure Code No. 35.

5.1.4 *Level B.* — Packaging and packing shall conform to the requirements of Military Specification MIL–P–12902, Procedure Code No. 21.

5.1.5 *Level C.* — The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 **Marking—Nonmilitary.** — In addition to any special marking shown in the contract or order, packages and exterior containers shall be marked in accordance with Federal Standard No. 123.

5.3 **Marking—Military.** — Marking shall conform to the requirements of Military Specification MIL–P–12902, except that the ASMPA stock number will not be required.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Types required (see 1.2).

(*c*) Interchangeable removable tip, when required (see 3.2.3.2 and figure 3).

(*d*) Selection of applicable levels of packaging and packing (see 5.1).

(1) Level B packaging and packing is intended to provide economical but limited protection, and should be specified only when it is determined the cannulae will be held in covered storage not more than 12 months from date of initial packaging.

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 **Transportation description.**—Transportation description applicable to this item is:

Rail:

Surgical Instruments, NOIBN, in Boxes.
Carload minimum weight 10,000 pounds subject to Rule 34 of Uniform Freight Classification.

Motor:

Surgical Instruments, NOI.
Truckload minimum weight (w) 10.6 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force.

PREPARING ACTIVITY:
Navy—Bureau of Medicine and Surgery (MMSA).

Copies of this specification may be purchased for 5 cents each.

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